module-14-magic-squares - Darshana (60-min workshop)

IKS Curriculum - senior band

Module 14 — Magic Squares — Bhadragaṇita (Darśana)

Sanskrit: *Bhadragaṇita* · **Band:** Senior (9-12) · **Duration:** 60 min · **Path:** Darśana (Workshop)

Nārāyaṇa Paṇḍita's 1356 CE magic-square algorithms.

> **Darśana note.** This workshop condenses Module 14's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to **Adhyayana** (the Full Course path).

Opening hook · 5 min

Project the Khajuraho 4×4 magic square (~10th c. CE) and Dürer's Melencolia I 4×4 (1514). *"Same combinatorial object. Today: Nārāyaṇa Paṇḍita's 1356 CE algorithm — most rigorous pre-modern treatment of magic squares."*

Core idea · 15 min

Magic squares — combinatorics + history.

Algebra: - Magic constant $M = n(n^2 + 1) / 2$. - For $n = 3$: 15. For $n = 4$: 34. For $n = 5$: 65. - Number of essentially distinct 3 × 3 magic squares (using 1-9, up to symmetry): EXACTLY 1. - 4 × 4: 880 (Frenicle de Bessy, 1693). - 5 × 5: 275,305,224.

Nārāyaṇa Paṇḍita's 1356 stair-step algorithm: - Place 1 in top-middle (or any specified start). - Move up-right; wrap with toroidal topology. - If destination occupied, move down instead.

Honest historical framing: - Lo Shu (China, ~9th c. BCE): earliest known 3 × 3 magic square. - Khajuraho (~10th c. CE): pan-diagonal 4 × 4 — sophisticated combinatorial property. - Gaṇita-kaumudī (1356 CE): rigorous algorithmic treatment. - Dürer (1514): European 4 × 4.

The mathematics is universal; the algorithmic systematisation by Nārāyaṇa is a genuine contribution to combinatorics.

Demonstration · 10 min

Compare three magic squares. Khajuraho (4 × 4, pan-diagonal), Dürer (4 × 4, “ordinary”), and a stair-step 5 × 5. Identify what makes Khajuraho special.

Source moment · 5 min

Quoted reference. Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* (1356 CE), *Bhadragaṇita* chapter. The treatise systematises magic-square construction for any odd order. Translated and analysed by Kim Plofker in *Mathematics in India* (2009). Predates the most rigorous European treatments (Stifel 1544, Fermat 1640) by ~200 years.

Hands-on activity · 15 min

Pan-diagonal 4 × 4 challenge. Construct a 4 × 4 magic square where ALL diagonals (including “broken” diagonals that wrap around) sum to 34. (Hint: the Khajuraho square is one such.) Time it.

Reflection + take-home · 10 min

“Nārāyaṇa’s 1356 work predates Stifel by ~200 years. Why is it absent from most Western histories of mathematics? Discuss the politics of mathematical historiography.”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — Discover the 3 × 3
- **Day 02** — Magic constant
- **Day 03** — How many 3 × 3 magic squares?
- **Day 04** — Stair-step method
- **Day 05** — Practice — 5 × 5 and 7 × 7
- **Day 06** — 4 × 4 and the Khajuraho square
- **Day 07** — Historical context — Nārāyaṇa Paṇḍita
- **Day 08** — Cross-cultural — Lo Shu, Dürer, Indian
- **Day 09** — Design your own magic square
- **Day 10** — Final share + assessment

→ **Open the Adhyayana track for this module** to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.